Unit Absorption Asymmetry at Prescribed Extinction for Compact Reciprocal Dielectrics

Abstract

For two opposite real linearly polarized plane waves, reciprocity fixes the extinction of a compact scatterer but permits different absorption and scattering. At a fixed vacuum wavenumber $k > 0$, we establish that the supremum of the absorption difference divided by common extinction is one at every exactly prescribed extinction $E_* > 0$. The realization uses finite compact two-layer local anisotropic nonmagnetic dielectrics satisfying arbitrary fixed upper caps $P > 1$ and $Q > 0$, with $I \preceq \mathrm{Re}_H\, \varepsilon \preceq PI$ and $0 \prec \mathrm{Im}_H\, \varepsilon \preceq QI$ in each layer. Every individual body has asymmetry strictly below one. The constructive step is a physical aperture limit uniform after factorization by the squared sine of a material rotation angle, including its zero-angle auxiliary limit. This permits the coupling angle to decrease as the aperture increases while total extinction remains fixed. A subsequent positive isotropic-loss perturbation and a second intermediate-value argument preserve the exact target. Geometric resources are unrestricted across the approximating sequence, and no common positive loss floor or bandwidth is imposed. The polarization mechanism and reciprocal extinction identity have prior precedents; the contribution is the controlled compact Maxwell realization at prescribed extinction. A frozen Lean 4 formalization certifies the stated existence, suprema, and physical power identities, with the scope of its verification distinguished from independent semantic and priority review.

1 Introduction and relation to OQ7

Open Question 7 in the tutorial of Miller and Monticone asks for fundamental limits on the redistribution between absorption and scattering under time-reversed illumination [1]. A normalization, illumination class, and resource class are needed to turn that question into a theorem. Here the normalization is the common dimensional plane-wave extinction, the illumination pair is opposite real linear polarization in vacuum, and the objects are local reciprocal dielectric bodies at one frequency.

The result is an existence and supremum theorem. It states that a finite object can have exactly any prescribed positive extinction and an absorption difference arbitrarily close to that extinction, while its real permittivity and loss obey fixed pointwise upper bounds. It establishes neither a finite-resource optimum nor finite-bandwidth performance. The same material tensor is used in both experiments, and scattering includes the whole outgoing sphere and both polarizations.

At a fixed coupling angle, enlarging a planar-like aperture increases extinction proportionally to area. A fixed-stack aperture limit therefore does not suffice to prescribe total extinction. The exact rotation factorization and uniform normalized aperture theorem below permit the coupling angle to decrease while the aperture grows. Green-operator and localization arguments justify this passage for actual finite Maxwell bodies. Dipole optimization, scalar thickness laws, fixed-resource bounds, and quartic small-size results have different hypotheses and are outside the present proof.

2 Prior work and the precise contribution

Reciprocal extinction equality and directional absorption are established facts. Sounas and Alù derived opposite-incidence extinction symmetry and its implications for scattering manipulation [2]. Hopkins and coauthors studied directional absorption and scattering at equal extinction through modal interference in reciprocal particles [3]. The passive inequality used below is a consequence of power conservation, and its strictness follows from the optical theorem. We claim no independent discovery of these facts.

Polarization conversion combined with selective attenuation also has a substantial history. Armitage gives reciprocity constraints for Jones matrices and a polarizer–waveplate example [4]. Khurgin and coauthors demonstrated reciprocal asymmetric polarization switching using selective attenuation and imperfect waveplates, including passive experiments [5]. This is a closely related channel mechanism. Ra'di, Asadchy, and Tretyakov analyze thin bianisotropic absorbers and distinct opposite-side responses, distinguishing reciprocal and nonreciprocal constitutive classes [6]. Such sheet and channel constructions establish relevant mechanisms; their realizability assumptions must be retained when comparing them with local nonmagnetic dielectric volumes. Likewise, the lossless-response time-reversal bounds of Sounas and Alù concern a different set of hypotheses [7].

Andersen and Frandsen showed that receiving-antenna absorption divided by absorption plus total scattering can approach one, with a directivity-dependent cost [8]. That result precedes any general near-unit absorption-efficiency claim. It does not by itself supply the present simultaneous opposite-incidence response at every prescribed extinction within a capped local dielectric class. Schab, Gustafsson, and collaborators formulate absorption–scattering tradeoffs with prescribed losses and material parameters [9]; those finite-resource optimization problems motivate the resource distinctions maintained here. Khurgin and Sun's loss-scaling analysis supplies separate material and confinement context for resonant metal–dielectric structures [10].

Related layered and metasurface devices also require explicit comparison. Kononchuk and coauthors combine dichroic attenuation with magnetic Faraday rotation in a layered-sheet isolator [11]; its complete device has a different, magnetically nonreciprocal constitutive class, while its selective-attenuation component is relevant. Earlier bianisotropic metasurface asymmetric absorption was studied by Yazdi and coauthors [12]. Máñez-Espina and coauthors derive maximum directional absorptance difference in a reciprocal bianisotropic metasurface model and verify its predictions with full-wave simulations [13]. A planar absorptance difference in a specified coupled-mode model is distinct from a compact dimensional extinction-normalized supremum with every positive target and arbitrary material caps. These comparisons do not establish first priority for the latter theorem.

For the finite-to-planar passage, Gao, Yang, and Kattawar studied polarized extinction of plates with large aspect ratios [14]. Gustafsson and coauthors derived planar extinction and scattered-transmission identities for periodic slabs [15]; literal compact-material truncation and the uniform normalized limit required here have different quantifiers. Windowed Green-function methods rigorously treat layered backgrounds [16, 17], while volume-integral formulations supply the singular-kernel and radiation framework [18, 19]. A computational window applied to an infinite-interface equation and literal truncation of a material body are different operations. Our smooth windows enter only trial currents; the final material aperture remains sharp.

Mann and Alù's 2023 conference abstract explicitly announces maximal directional absorption for bianisotropic dipolar particles [20]. It is direct prior art for the qualitative extremal phenomenon. The full equations, normalization, and any equivalent compact dielectric realization have not been verified from that contribution. Priority against it remains unresolved. More generally, absence of

an identical retrieved statement does not establish first priority.

The contribution established here is the combination of an aperture theorem uniform at vanishing physical coupling, exact finite-object extinction tuning, and positive-loss regularization preserving arbitrary fixed upper material caps. The mechanism, passive bound, extinction symmetry, and planar optical theorem have the prior precedents described above.

3 Maxwell model, material class, and cross sections

Use time dependence $e^{-i\omega t}$, vacuum wavenumber $k = \omega/c > 0$, relative permeability $\mu = I$, and relative permittivity $\varepsilon = I + \chi$. Write $\mathbf{H} = Z_0 H$ for the impedance-normalized magnetic field. The incident fields are

$$f_\pm(\mathbf{x}) = \mathbf{e}_y e^{\pm ikz}, \qquad \mathbf{H}_{\mathrm{inc},\pm} = \pm(\mathbf{e}_z \times \mathbf{e}_y) e^{\pm ikz}. \tag{1}$$

Their electric amplitudes are one and their incident intensities coincide. This is a time-reversed pair of vacuum waves. The absorbing object itself is reciprocal, without requiring its lossy dynamics to be invariant under time reversal.

Let $R, h, d > 0$, $S_R = (-R/2, R/2)^2$, and $D_R = S_R \times (0, h+d)$. The absorber occupies $S_R \times (0, h)$ and the plate occupies $S_R \times (h, h+d)$. Each layer has a constant complex 3×3 tensor. The coefficient is I outside D_R; boundary values do not affect the weak equations. The closure of the material region is compact. For fixed $P > 1, Q > 0$, both tensors satisfy

$$\varepsilon^T = \varepsilon, \qquad I \preceq \mathrm{Re}_H\, \varepsilon \preceq PI, \qquad 0 \prec \mathrm{Im}_H\, \varepsilon \preceq QI, \tag{2}$$

where $\mathrm{Re}_H\, T = (T + T^\dagger)/2$ and $\mathrm{Im}_H\, T = (T - T^\dagger)/(2i)$. Strict positivity holds separately for each selected finite object. The class imposes no common lower loss bound.

The total fields are locally square integrable and satisfy, distributionally,

$$\nabla \times E = ik\mathbf{H}, \qquad \nabla \times \mathbf{H} = -ik\varepsilon E. \tag{3}$$

The equations are tested against every C_c^∞ complex vector field with the bilinear distribution pairing. Thus the fields belong locally to $H(\mathrm{curl})$. The scattered fields $E_s = E - f_\pm$ and $\mathbf{H}_s = \mathbf{H} - \mathbf{H}_{\mathrm{inc},\pm}$ obey the uniform Silver–Müller condition

$$r\|\mathbf{H}_s(r\mathbf{n}) \times \mathbf{n} - E_s(r\mathbf{n})\| \longrightarrow 0 \quad (r \to \infty), \quad \text{uniformly for } \mathbf{n} \in S^2. \tag{4}$$

Weak fields are unique almost everywhere. Their physical far field and sphere-flux limits refer to the canonical radiating exterior representative, avoiding arbitrary pointwise changes on null sets.

Put $p = \chi E$ and use an inner product conjugate-linear in its first argument. For unit electric incident amplitude, the dimensional cross sections are

$$X_\pm = k\Im\langle f_\pm, p_\pm\rangle_{D_R}, \qquad A_\pm = k\int_{D_R} E_\pm^\dagger\, \mathrm{Im}_H\, \chi E_\pm \,\mathrm{d}\mathbf{x}. \tag{5}$$

The far-field amplitude and total scattering are

$$F_p(\mathbf{n}) = \frac{k^2}{4\pi}\Pi_{\mathbf{n}} \int_{D_R} p(\mathbf{x}) e^{-ik\mathbf{n}\cdot\mathbf{x}}\,\mathrm{d}\mathbf{x}, \quad \Pi_{\mathbf{n}} = I - \mathbf{n}\mathbf{n}^T, \qquad S = \int_{S^2} \|F_p(\mathbf{n})\|^2\,\mathrm{d}\Omega. \tag{6}$$

Compactly supported L^2 currents are L^1, so this far field is continuous. It is the actual radiation amplitude, with $E_s(r\mathbf{n}) = e^{ikr}F_p(\mathbf{n})/r + O(r^{-2})$ uniformly on the sphere. Reciprocity and the optical theorem give

$$X_+ = X_- = X, \qquad X = A_\pm + S_\pm, \qquad X_\pm = \frac{4\pi}{k}\Im\{\mathbf{e}_y^\dagger F_{p_\pm}(\pm\mathbf{e}_z)\}. \tag{7}$$

When $X > 0$, define

$$\Gamma = \frac{|A_+ - A_-|}{X} = \frac{|S_+ - S_-|}{X}. \tag{8}$$

X, A, S, E_* have units of area; Γ is dimensionless. Infinite plane waves are never assigned a finite total incident power. Section 5 establishes the physical identities used here.

4 Main theorem and nonattainment

Let $\mathcal{C}_{k,P,Q,E_*}$ consist of all two-layer square-aperture bodies just defined whose opposite-incidence solutions have common extinction $E_* > 0$.

Theorem 4.1 (Prescribed-extinction supremum)**.** *Fix $k > 0$, $P > 1$, $Q > 0$, and $E_* > 0$. For every $0 < \epsilon < 1$, there exists a body satisfying (2), with finite positive R, h, d, whose unique outgoing weak Maxwell solutions for (1) obey*

$$X_+ = X_- = E_* \quad \textit{exactly}, \qquad 1 - \epsilon < \Gamma < 1. \tag{9}$$

Consequently,

$$\sup_{\mathcal{C}_{k,P,Q,E_*}} \Gamma = 1, \qquad \sup_{\mathcal{C}_{k,P,Q,E_*}} |A_+ - A_-| = E_*. \tag{10}$$

Neither supremum is attained. No common bound on aperture, thickness, volume, integrated material resources, or positive loss floor is imposed across the approximating sequence.

Lemma 4.2 (Passive bound and strict nonattainment)**.** *Every individual body in the stated class with $X > 0$ has $0 \le \Gamma < 1$.*

Proof. Passivity and (7) give $0 \le A_\pm \le X$, hence $\Gamma \le 1$. Equality would require one experiment to have $A = X$ and $S = 0$. By continuity, a far field with zero integral of its squared norm vanishes in every direction. Its forward optical theorem then gives $X = 0$, a contradiction. This strictness argument also applies to any compact passive reciprocal object for which the same physical identities hold; the formal class of Theorem 4.1 is the two-layer class. □

The existence construction is proved in Sections 6–8. Lemma 4.2 supplies the upper bound and nonattainment; the constructed sequence supplies the lower bound on each supremum.

5 Exact volume operators and physical power identities

For any bounded D, extend $p \in L^2(D)^3$ by zero and define

$$K_D p = (k^2 I + \nabla\nabla\cdot)(g_k * p)\big|_D, \qquad g_k(\mathbf{x}) = \frac{e^{ik|\mathbf{x}|}}{4\pi|\mathbf{x}|}. \tag{11}$$

The derivatives are distributional, including the static contact term. In this section K^{stat} denotes the electrostatic operator; the later transverse fiber $K_{\boldsymbol{\kappa}=0}$ is a different operator at the nonzero frequency k.

Lemma 5.1 (Boundedness and radiation identity)**.** *The restricted operator K_D is bounded on $L^2(D)^3$, and*

$$\|K^{\mathrm{stat}}\| \le 1, \qquad \|K_D - K_D^{\mathrm{stat}}\| \le \frac{3}{4}k^2 \left(\frac{3|D|}{4\pi}\right)^{2/3}. \tag{12}$$

For every such current,

$$\Im\langle p, K_D p\rangle = \frac{k^3}{16\pi^2}\int_{S^2}\left\|\Pi_{\mathbf{n}}\int_D p(\mathbf{x})e^{-ik\mathbf{n}\cdot\mathbf{x}}\,\mathrm{d}\mathbf{x}\right\|^2\mathrm{d}\Omega \geq 0. \tag{13}$$

Proof. The Newtonian part has Fourier multiplier $-\boldsymbol{\xi}\boldsymbol{\xi}^T/|\boldsymbol{\xi}|^2$, with norm at most one. It includes the local distributional contribution. Set $b(u) = e^{iu}(iu-1)+1$. Since $b'(u) = -ue^{iu}$ and $b(0) = 0$, $|b(u)| \leq u^2/2$ for real $u \geq 0$. The dynamic difference kernel has radial and tangential eigenvalues $-2b(kr)/(4\pi r^3)$ and $((kr)^2e^{ikr} + b(kr))/(4\pi r^3)$, respectively. Its operator norm is at most $3k^2/(8\pi r)$. Rearrangement of the radial decreasing kernel gives $\sup_{\mathbf{x}}\int_D|\mathbf{x}-\mathbf{y}|^{-1}\,\mathrm{d}\mathbf{y} \leq 2\pi(3|D|/(4\pi))^{2/3}$. Schur's test proves (12).

The smooth identity $\Im g_k(\mathbf{x}) = k(16\pi^2)^{-1}\int_{S^2}e^{ik\mathbf{n}\cdot\mathbf{x}}\,\mathrm{d}\Omega$ follows by direct angular integration. Applying $k^2I + \nabla\nabla\cdot$ supplies $k^2\Pi_{\mathbf{n}}$. Reciprocity of the kernel, Fubini's theorem, and $L^2(D)$ density yield (13). The static contact term is real and contributes no imaginary work. The large-distance expansion of the same kernel gives (6) and its uniform remainder, together with $\mathbf{H}_s(r\mathbf{n}) = e^{ikr}\mathbf{n}\times F_p(\mathbf{n})/r + O(r^{-2})$. Their sphere Poynting flux tends to S. □

Proposition 5.2 (Weak Maxwell well-posedness and optical identities)**.** *For every body satisfying* (2)*, each incidence has a unique outgoing weak solution up to equality almost everywhere. Its powers satisfy* (7) *and* $A_\pm, S_\pm \geq 0$.

Proof. For this fixed body, $\operatorname{Im}_H\chi \succeq mI$ and $\|\chi\| \leq M$ for some $m, M > 0$, because there are two constant tensors. Hence χ is invertible and

$$W = -\operatorname{Im}_H\chi^{-1} = \chi^{-\dagger}(\operatorname{Im}_H\chi)\chi^{-1} \succeq \frac{m}{M^2}I. \tag{14}$$

Set $L_D = \chi^{-1} - K_D$. Then $-\Im\langle p, L_D p\rangle \geq (m/M^2)\|p\|^2$ by Lemma 5.1. Complex Lax–Milgram applied to $i\langle q, L_D p\rangle$ gives a unique solution of $L_D p = f$, with inverse norm at most M^2/m. Reconstruct $E = f + (k^2I + \nabla\nabla\cdot)(g_k * p)$ and $\mathbf{H} = (ik)^{-1}\nabla\times E$. Local H^2 regularity of the scalar volume potential gives locally square-integrable fields and distributional Maxwell equations; their scattered fields are outgoing. Conversely, subtracting this reconstruction from any outgoing weak state produces an entire outgoing vacuum Maxwell pair, which vanishes almost everywhere by Appendix A. Thus the volume and weak formulations describe the same solution.

Inside D, $E = \chi^{-1}p$, so $A = k\langle p, Wp\rangle$. Taking the imaginary part of $\langle p, L_D p\rangle = \langle p, f\rangle$ gives

$$k\Im\langle f, p\rangle = k\langle p, Wp\rangle + k\Im\langle p, K_D p\rangle = A + S. \tag{15}$$

The last equality is (13). Substitution of (6) at the forward direction proves the forward optical theorem. For reciprocity, use $[u, v] = \int_D u^T v$ without conjugation. Symmetry of χ^{-1} and of the distributional Green kernel gives $[u, L_D^{-1}v] = [v, L_D^{-1}u]$. Since $f_- = \overline{f_+}$, $\langle f_+, L_D^{-1}f_+\rangle = [f_-, L_D^{-1}f_+] = [f_+, L_D^{-1}f_-] = \langle f_-, L_D^{-1}f_-\rangle$. Their imaginary parts coincide. This proves all stated identities for the actual weak solutions. □

6 Planar dielectric construction and rotation factorization

Initially use passive rank-one susceptibilities

$$\chi_A = i\eta\mathbf{e}_x\mathbf{e}_x^T, \qquad \chi_{W,\theta} = (a+i\delta)\mathbf{u}_\theta\mathbf{u}_\theta^T, \qquad \mathbf{u}_\theta = \cos\theta\mathbf{e}_x + \sin\theta\mathbf{e}_y, \tag{16}$$

with $\eta, a, \delta > 0$. These intermediate tensors have loss-free null directions and are not the final admissible tensors. Their local rank-one law reduces the unknown current to a scalar, while propagation and radiation remain the full vector Maxwell problem. Figure 1 summarizes the geometry and polarization axes.

Let $J_\theta : L^2(D_R) \to L^2(D_R)^3$ multiply a scalar by $\mathbf{e}_x$ in the absorber and by $\mathbf{u}_\theta$ in the plate. It is a real isometry. Let $\alpha(z) = i\eta$ below h and $\alpha(z) = a + i\delta$ above h. The scalar operator is

$$L_{R,\theta} = \alpha^{-1} - J_\theta^* K_{D_R} J_\theta, \qquad L_{R,\theta} q = J_\theta^* f. \tag{17}$$

Its loss weight and coercivity constant are

$$w = -\Im \alpha^{-1}, \qquad c_0 = \min\left\{\eta^{-1}, \frac{\delta}{a^2 + \delta^2}\right\} > 0. \tag{18}$$

Lemma 5.1 gives $-\Im\langle q, L_{R,\theta} q\rangle \geq c_0 \|q\|^2$. Lax–Milgram proves

$$\|L_{R,\theta}^{-1}\| \leq c_0^{-1}, \tag{19}$$

independently of aperture and angle. The same reconstruction and converse argument as above establishes uniqueness for these intermediate rank-one bodies: every physical current lies in the range of J_θ, so every weak solution satisfies (17).

For (1), $J_\theta^* f_\pm = \sin\theta\, g_\pm$, where $g_\pm = \mathbf{1}_{(h,h+d)}(z) e^{\pm ikz}$. Define

$$r_{R,\theta}^\pm = L_{R,\theta}^{-1} g_\pm, \qquad p_{R,\theta}^\pm = \sin\theta\, J_\theta r_{R,\theta}^\pm. \tag{20}$$

The auxiliary $r^\pm$ are defined even at $\theta = 0$, where the physical y illumination induces no current. The exact physical powers factor as

$$\begin{aligned} X_{R,\theta}^\pm &= \sin^2\theta\, Y_R(\theta), & Y_R^\pm &= k\Im\langle g_\pm, r_{R,\theta}^\pm\rangle, \\ A_{R,\theta}^\pm &= \sin^2\theta\, B_R^\pm(\theta), & B_R^\pm &= k\langle r_{R,\theta}^\pm, w r_{R,\theta}^\pm\rangle, \\ S_{R,\theta}^\pm &= \sin^2\theta\, C_R^\pm(\theta), & C_R^\pm &= k\Im\langle J_\theta r_{R,\theta}^\pm, K_{D_R} J_\theta r_{R,\theta}^\pm\rangle. \end{aligned} \tag{21}$$

Bilinear reciprocity with $g_- = \overline{g_+}$ gives $Y_R^+ = Y_R^- = Y_R > 0$. Also $Y_R = B_R^\pm + C_R^\pm$, and for $\sin\theta \neq 0$,

$$\Gamma_R(\theta) = \frac{|B_R^+(\theta) - B_R^-(\theta)|}{Y_R(\theta)}. \tag{22}$$

Only a physical material angle changes; the incident intensity stays fixed.

6.1 A reflectionless plate under any upper real-permittivity cap

Given $P > 1$, choose an integer $j \geq 1$ such that

$$p_j = \left(\frac{2j}{2j-1}\right)^2 \leq P, \qquad a = p_j - 1, \qquad kd = (2j-1)\pi. \tag{23}$$

At $\delta = 0$, the phase along $\mathbf{u}_\theta$ is $2j\pi$, and the orthogonal transverse phase is $(2j-1)\pi$. Matching tangential $E, \mathbf{H}$, equivalently the scalar E, E', gives zero reflection in both principal directions. The boundary-plane Jones transmission and the vacuum plate phase are

$$H_\theta = 2\mathbf{u}_\theta \mathbf{u}_\theta^T - I_2, \qquad v_W = e^{ikd} = -1. \tag{24}$$

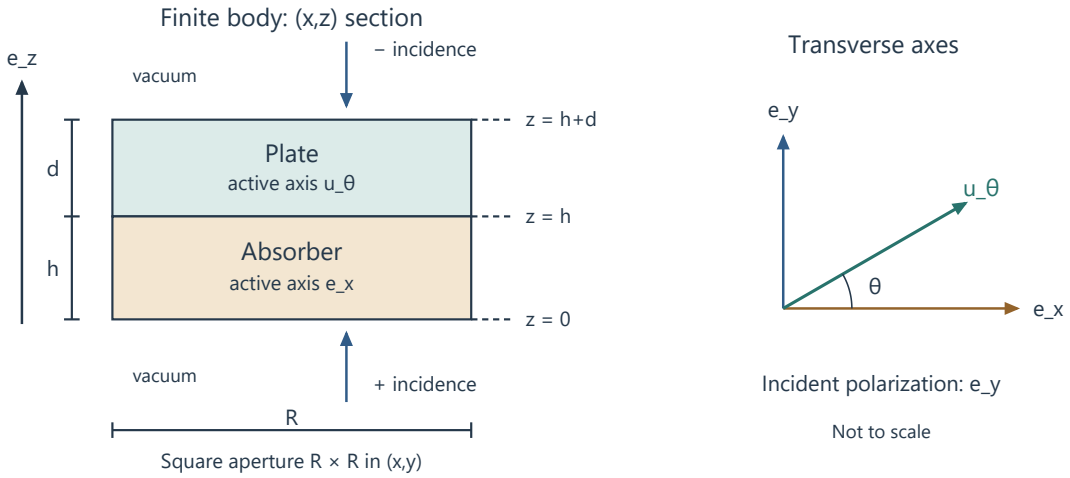


Figure 1: Finite two-layer geometry and transverse material axes, not to scale. Both incident waves have electric polarization $\mathbf{e}_y$. The $+$ wave encounters the absorber first; the $-$ wave encounters the plate first. Each layer has square transverse aperture $R \times R$. The susceptibilities in (16) are intermediate; adding $i\nu I$ in (42) gives the final strictly lossy tensors.

Indeed, propagation of $(E, E'/k)^T$ through a scalar layer of index n is the matrix with entries $\cos(nkd), \sin(nkd)/n, -n\sin(nkd), \cos(nkd)$; at an integer multiple of π it is $\pm I_2$.

All slab coefficients are referenced to their boundary planes: unit incident electric amplitude at one interface gives reflected amplitude r there and transmitted amplitude t at the other interface. A vacuum layer of width ℓ has $r = 0, t = e^{ik\ell}$. For the absorber's x polarization, put $n = \sqrt{1 + i\eta}$ on the passive branch, $r_0 = (1-n)/(1+n)$, and $p = e^{inkh}$. Its exact amplitudes and absorptance are

$$r_A = \frac{r_0(1-p^2)}{1-r_0^2p^2}, \qquad t_A = \frac{(1-r_0^2)p}{1-r_0^2p^2}, \qquad A_x = 1 - |r_A|^2 - |t_A|^2. \tag{25}$$

$|r_0| < 1$ and $|p| < 1$ make the denominator nonzero and $t_A \neq 0$. The scalar flux identity $A_x = k\eta \int_0^h |E_x|^2 \, \mathrm{d}z$ proves $A_x > 0$: a zero field inside would force both $1 + r_A = 0$ and $1 - r_A = 0$ at the incident interface. Thus $0 < A_x < 1$.

Lemma 6.1 (Regular normalized planar limit)**.** *At zero plate loss and* $\sin\theta \neq 0$*, the physical planar power densities are*

$$X_\infty = 4\sin^2\theta, \qquad A_\infty^+ = 0, \qquad A_\infty^- = 4\sin^2\theta\cos^2\theta A_x. \tag{26}$$

Their normalized values extend to $\theta = 0$ *with* $Y_\infty = 4$, $B_\infty^+ = 0$, $B_\infty^- = 4\cos^2\theta A_x$. *For fixed finite* h, η, j, *the normalized powers are jointly continuous in* (δ, θ) *near* $(0, 0)$, *and*

$$Y_{\infty,\delta}(0) \to 4, \qquad \gamma_{\infty,\delta}(0) := \frac{|B_{\infty,\delta}^+(0) - B_{\infty,\delta}^-(0)|}{Y_{\infty,\delta}(0)} \to A_x \quad (\delta \downarrow 0). \tag{27}$$

Proof. From the absorber side, y crosses vacuum-equivalent material before the reflectionless plate. From the plate side, $H_\theta \mathbf{e}_y$ has x component $\sin 2\theta$, giving absorption $\sin^2 2\theta A_x$. In both directions the co-polar transmission relative to vacuum is $\cos 2\theta$. The planar optical theorem is $X_\infty = 2(1 - \Re t_{\mathrm{co}})$, with scattered transmission defined by subtracting the vacuum wave [15]. This proves (26).

Let r_W, t_W be scalar slab coefficients for permittivity $p_j + i\delta$ and width d. Put $D = 1 - r_A r_W \cos^2\theta$. Define t_{co} as the complete-stack co-polar transmitted amplitude divided by the vacuum factor $e^{ik(h+d)} = v_A v_W$. Exact cascade matching gives

$$t_{\mathrm{co}} = 1 + \frac{(t_W/v_W - 1)\sin^2\theta}{D}, \qquad Y_{\infty,\delta}(\theta) = 2\Re\left\{\frac{1 - t_W/v_W}{D}\right\}. \tag{28}$$

At $(\delta, \theta) = (0, 0)$, $r_W = 0, t_W = 1, D = 1$. Appendix B supplies every scattered amplitude divided by $\sin\theta$ as a nonsingular continuous expression. Squared norms give the continuous normalized scattering, and $B_\infty = Y_\infty - C_\infty$ gives continuous absorption. This proves (27) without dividing two vanishing quantities. □

Finally take $\eta = 1/N$ and $kh = N^2$. Then $r_0 \to 0$, while $|p| = \exp[-N^2\Im\sqrt{1 + i/N}] \to 0$, since $N\Im\sqrt{1 + i/N} \to 1/2$. Hence $A_x \to 1$. Every selected h, η is finite and positive; j stays fixed for the cap P.

7 Uniform physical-aperture estimates

Write $L = h + d$ and $\mathcal{H} = L^2((0, L))$. The planar scalar operator uses the same J_θ, α, w, now on the interval. Let $K_{\boldsymbol{\kappa}}$ be the transverse Fourier fiber of the vector Green operator, and set

$$B_{\boldsymbol{\kappa},\theta} = J_\theta^* K_{\boldsymbol{\kappa}} J_\theta, \qquad r^{\pm}_{\infty,\theta} = (\alpha^{-1} - B_{\mathbf{0},\theta})^{-1} g_\pm. \tag{29}$$

The planar $Y_\infty, B^{\pm}_\infty, C^{\pm}_\infty$ are given by the normalized work formulas on $(0, L)$, per unit area. They are dimensionless; Y_R, B_R, C_R have units of area. The auxiliary planar response at angle zero is not the physical zero-angle illumination.

Theorem 7.1 (Uniform normalized aperture limit)**.** *Fix $k, h, d, \eta, a, \delta > 0$ and a compact angle interval I_θ. Extend $r^{\pm}_{\infty,\theta}$ constantly in the transverse coordinates. As $R \to \infty$,*

$$\sup_{\theta \in I_\theta} \frac{\|r^{\pm}_{R,\theta} - r^{\pm}_{\infty,\theta}\|_{L^2(D_R)}}{R} \longrightarrow 0. \tag{30}$$

Uniformly on I_θ,

$$\frac{Y_R}{R^2} \to Y_\infty, \qquad \frac{B^{\pm}_R}{R^2} \to B^{\pm}_\infty, \qquad \frac{C^{\pm}_R}{R^2} \to C^{\pm}_\infty. \tag{31}$$

The auxiliary ratios converge uniformly as well. These statements include arbitrary sequences $\theta_R \to 0$ in I_θ. Constants may depend on the fixed thicknesses and positive active losses; no joint uniformity as those losses vanish or thicknesses diverge is asserted.

7.1 Weyl fibers and the grazing singularity

Use $\widehat{v}(\boldsymbol{\kappa}) = \int_{\mathbb{R}^2} e^{-i\boldsymbol{\kappa}\cdot\mathbf{x}} v(\mathbf{x})\,\mathrm{d}\mathbf{x}$, with inverse factor $(2\pi)^{-2}$. Let

$$\beta(\boldsymbol{\kappa}) = \sqrt{k^2 - |\boldsymbol{\kappa}|^2}, \qquad G_{\boldsymbol{\kappa}}(z) = \frac{i}{2\beta(\boldsymbol{\kappa})} e^{i\beta(\boldsymbol{\kappa})|z|}, \tag{32}$$

on the outgoing/decaying branch. For transverse indices $a, b \in \{1, 2\}$,

$$\begin{aligned} (K_{\boldsymbol{\kappa}})_{ab} &= (k^2\delta_{ab} - \kappa_a\kappa_b) G_{\boldsymbol{\kappa}} *, \\ (K_{\boldsymbol{\kappa}})_{a3} &= (K_{\boldsymbol{\kappa}})_{3a} = i\kappa_a \partial_z G_{\boldsymbol{\kappa}} *, \\ (K_{\boldsymbol{\kappa}})_{33} &= |\boldsymbol{\kappa}|^2 G_{\boldsymbol{\kappa}} * - I. \end{aligned} \tag{33}$$

Convolution integrates the source height over $(0, L)$. These formulas follow by transverse Fourier transformation of $(-\Delta - k^2)g_k = \delta_0$. In particular, $\partial_z^2 G_{\boldsymbol{\kappa}} = -\beta^2 G_{\boldsymbol{\kappa}} - \delta_0$ supplies the longitudinal local term.

Since $|G| \le (2|\beta|)^{-1}$ and $|\partial_z G| \le 1/2$ off the diagonal, Schur estimates yield finite constants depending on k, L such that

$$\begin{aligned} \|K_{\boldsymbol{\kappa}}\| &\le C\left(1 + |\boldsymbol{\kappa}| + \frac{k^2 + |\boldsymbol{\kappa}|^2}{|\beta(\boldsymbol{\kappa})|}\right), \\ \|K_{\boldsymbol{\kappa}} - K_{\mathbf{0}}\| &\le C_1|\boldsymbol{\kappa}|, \qquad |\boldsymbol{\kappa}| \le k/2. \end{aligned} \tag{34}$$

The second bound follows from bounded derivatives of the kernels on the low-frequency disk. The possible grazing singularity is $\big||\boldsymbol{\kappa}| - k\big|^{-1/2}$, locally integrable in two dimensions; its square need not be integrable. At zero transverse wavevector the active transverse kernel is $(ik/2)e^{ik|z-z'|}$ times the dot product of the active axes. Its imaginary quadratic form is a positive sum of the two outgoing planar fluxes. Thus the planar inverse has the same bound c_0^{-1} and $\|r^{\pm}_{\infty,\theta}\| \le c_0^{-1}\|g_\pm\|$ uniformly in angle.

7.2 Mixed Fourier localization proof

Proof of Theorem 7.1. Choose a fixed $v \in C_c^\infty((-1/2, 1/2)^2)$ and put $v_R(\mathbf{x}) = v(\mathbf{x}/R)$. Extend J_θ constantly in the transverse coordinates on $\mathbb{R}^2 \times (0, L)$. For a compactly supported strip current p, define the unrestricted observation field $\mathcal{K}p = (k^2 I + \nabla\nabla\cdot)(g_k * \widetilde{p})$, where $\widetilde{p}$ is its zero extension in three dimensions. For $r \in \mathcal{H}$, set $\widetilde{H}_{R,\theta} = J_\theta^* \mathcal{K}(J_\theta(v_R r)) - v_R B_{\mathbf{0},\theta} r$ on this strip and $H_{R,\theta} = \widetilde{H}_{R,\theta}|_{D_R}$. The latter equals $J_\theta^* K_{D_R} J_\theta(v_R r) - v_R B_{\mathbf{0},\theta} r$ on D_R. The global auxiliary field has transverse Fourier representation

$$\widehat{\widetilde{H}}_{R,\theta}(\boldsymbol{\kappa}) = R^2 \widehat{v}(R\boldsymbol{\kappa})(B_{\boldsymbol{\kappa},\theta} - B_{\mathbf{0},\theta}) r. \tag{35}$$

First derive this identity for smooth compactly supported height profiles by the distributional Weyl formula. The transverse transform acts on the global auxiliary field before aperture restriction. Density extends the distributional identity to $\mathcal{H}$ and the following estimates to its aperture restriction.

For $|\boldsymbol{\kappa}| \le k/2$, Hilbert-valued Plancherel and (34) bound the global L^2 norm of the low-frequency part of $\widetilde{H}_{R,\theta}$ by $C\|r\|\|\nabla v_R\|_2 = C\|r\|\|\nabla v\|_2$. For the complementary region choose an integer $N > 3$. Schwartz decay gives $R^2|\widehat{v}(R\boldsymbol{\kappa})| \le C_{v,N} R^{2-N}|\boldsymbol{\kappa}|^{-N}$, and

$$\int_{|\boldsymbol{\kappa}|>k/2} |\boldsymbol{\kappa}|^{-N}(\|K_{\boldsymbol{\kappa}}\| + \|K_{\mathbf{0}}\|)\,\mathrm{d}\boldsymbol{\kappa} < \infty. \tag{36}$$

The grazing part is locally L^1; at infinity the radial integrand is $O(\rho^{2-N})$. Inverse Fourier L^1 control gives the complementary part of $\widetilde{H}_{R,\theta}$ a transverse $L^\infty(\mathcal{H})$ bound $C_{v,N} R^{2-N}\|r\|$. Restricting to the aperture gives an $L^2(D_R)$ bound $C_{v,N} R^{3-N}\|r\|$. Therefore

$$\|H_{R,\theta}\|_{L^2(D_R)} \le C_v\|r\| + o(1)\|r\| \tag{37}$$

uniformly in angle, because each J_θ is an isometry. Only the low-frequency part used global Plancherel. For the density passage, boundedness of the restricted K_{D_R} identifies the local L^2 limit with the volume field; the high-frequency inverse transforms converge in transverse $L^\infty(\mathcal{H})$. Thus no square of the grazing singularity is introduced.

Take $r = r^{\pm}_{\infty,\theta}$. The trial current has residual $(v_R - 1)g_{\pm} - H_{R,\theta}$ in (17). Apply (19) and compare $v_R r$ with the constant transverse extension of r:

$$\sup_{\theta \in I_\theta} \frac{\|r^{\pm}_{R,\theta} - r^{\pm}_{\infty,\theta}\|_{L^2(D_R)}}{R} \le C\|1 - v\|_{L^2((-1/2,1/2)^2)} + \frac{C_v}{R} + o(R^{-1}). \tag{38}$$

At fixed v, let $R \to \infty$; then approximate one in L^2 by such smooth windows. The left side is independent of the window, proving (30). The material aperture and incident field were never smoothed.

Both exact and planar-extended currents have norm $O(R)$ uniformly in angle. Cauchy–Schwarz in the work formula gives $|Y_R - R^2 Y_\infty| = o(R^2)$. Boundedness of w and the quadratic-form difference estimate give $|B^{\pm}_R - R^2 B^{\pm}_\infty| = o(R^2)$. Exact energy balance gives the scattering limit by subtraction. This controls the full integrated scattering, including edge radiation and a narrowing forward peak.

The zero-wavevector operator varies continuously in norm with θ; its uniformly bounded inverse makes the planar powers continuous. Positive w and nonzero $g_{\pm}$ imply $Y_\infty > 0$, so its minimum on I_θ is positive. Division therefore gives uniform convergence of the auxiliary ratios, including the zero-angle limit. □

8 Exact prescribed-extinction realization

8.1 A root in a finite rank-one object

Fix $0 < \epsilon < 1$. Choose a finite absorber with $0 < \eta < Q/3$ and $A_x > 1 - \epsilon/4$. Lemma 6.1 then permits a fixed $0 < \delta < Q/3$ such that $\gamma_{\infty,\delta}(0) > 1 - \epsilon/2$. Choose $\theta_0 > 0$ so that the planar normalized ratio exceeds $1 - 5\epsilon/8$ on $[0, \theta_0]$. Freeze these material parameters and thicknesses before enlarging the aperture.

Set $Y_0 = Y_{\infty,\delta}(0) > 0$ and define

$$\theta_{\mathrm{lo}}(R) = \sqrt{\frac{E_*}{2R^2 Y_0}}, \qquad \theta_{\mathrm{hi}}(R) = \sqrt{\frac{2E_*}{R^2 Y_0}}. \tag{39}$$

For sufficiently large finite R this bracket lies in $(0, \theta_0]$, and the uniform aperture theorem and the exact factorization imply

$$X_R(\theta_{\mathrm{lo}}(R)) \to E_*/2, \qquad X_R(\theta_{\mathrm{hi}}(R)) \to 2E_*. \tag{40}$$

Indeed, $Y_R(\theta)/R^2 \to Y_0$ along either endpoint and $\sin\theta/\theta \to 1$. For large R, throughout the bracket $\Gamma_R > 1 - 3\epsilon/4$.

At fixed R, J_θ and $L_{R,\theta}$ vary continuously in operator norm; the inverse identity and (19) give continuity of X_R. The intermediate value theorem supplies an angle θ_R with $X_R(\theta_R) = E_*$ exactly. Reciprocity supplies the same extinction for the other incidence. Monotonicity and a nonzero derivative are unnecessary. For every choice of such roots,

$$R\theta_R \to \sqrt{E_*/Y_0}, \qquad \Gamma_R(\theta_R) \to \gamma_{\infty,\delta}(0). \tag{41}$$

This scaling is an auxiliary statement about the intermediate rank-one family; it is not a uniform loss-regularized resource estimate.

8.2 Strict tensor loss without losing the exact target

For the selected finite R, add common isotropic loss:

$$\chi_{A,\nu} = i\eta \mathbf{e}_x \mathbf{e}_x^T + i\nu I, \qquad \chi_{W,\theta,\nu} = (a + i\delta)\mathbf{u}_\theta \mathbf{u}_\theta^T + i\nu I, \qquad \nu > 0. \tag{42}$$

Continuity at the rank-deficient tensor follows from an explicit full-vector electric-field inverse:

$$(I - K_{D_R} J_\theta \alpha J_\theta^*)^{-1} = I + K_{D_R} J_\theta L_{R,\theta}^{-1} J_\theta^*. \tag{43}$$

Multiplication in either order verifies this using $L_{R,\theta} = \alpha^{-1} - J_\theta^* K_{D_R} J_\theta$. The inverse norm is at most $1 + \|K_{D_R}\|/c_0$, uniformly on the compact angle bracket for this fixed body.

Adding $i\nu I$ changes the electric-field operator by $-i\nu K_{D_R}$. If $\nu\|K_{D_R}\|(1 + \|K_{D_R}\|/c_0) < 1$, a Neumann series gives an inverse converging uniformly to (43) as $\nu \downarrow 0$. Fields, currents, and their finite-domain power forms are consequently continuous in (ν, θ) uniformly on the bracket. Its unperturbed extinction has a positive minimum, so the normalized ratio is uniformly continuous there too.

Choose $0 < \nu < Q/3$ small enough to preserve both strict endpoint inequalities around E_* and $\Gamma > 1 - \epsilon$ everywhere in the bracket. A second intermediate-value argument gives a possibly adjusted angle with $X_+ = X_- = E_*$ exactly. In the absorber, the real eigenvalues are all one and the loss eigenvalues are $\eta + \nu, \nu, \nu$. In the plate the real eigenvalues are $p_j, 1, 1$ and the loss eigenvalues are $\delta + \nu, \nu, \nu$. They obey (2). The finite object has strictly positive tensor loss and satisfies (9); Lemma 4.2 supplies its strict upper bound. Letting $\epsilon \downarrow 0$ proves both suprema in (10).

The order of choices matters: choose a finite absorber, choose positive active plate loss, fix an angle interval, enlarge the aperture, and only then choose isotropic loss and retune. None of the estimates is asserted uniform in a vanishing active loss or a diverging thickness. The main theorem needs preservation of the bracket, not a quantitative rate for the final angle.

9 Physical consequences and limitations

At fixed extinction, let h denote the more absorbing incidence and ℓ the other one. Energy balance gives

$$1 - \Gamma = \frac{S_h + A_\ell}{E_*}. \tag{44}$$

Thus $\Gamma \to 1$ implies $A_h \to E_*$, $S_h \to 0$, $A_\ell \to 0$, and $S_\ell \to E_*$. No finite member has zero S_h at positive extinction. Define co-polar forward directivity by

$$D_{\mathrm{co},h} = \frac{4\pi |\mathbf{e}_y^\dagger F_h(\mathbf{d}_h)|^2}{S_h}, \qquad \Gamma \le 1 - \frac{k^2 E_*}{4\pi D_{\mathrm{co},h}}, \qquad \mathbf{d}_h = \pm \mathbf{e}_z. \tag{45}$$

The inequality follows by squaring the forward optical theorem and using $|\Im z| \le |z|$. It is an established efficiency/directivity obstruction, consistent with the antenna literature [8]. The expanding aperture permits the required unbounded directivity and a narrowing forward peak.

The selected positive principal-axis susceptibilities are compatible with passive causal scalar dispersion at the chosen frequency. For a target $a + ib$ with $a \ge 0, b > 0$, choose any $\gamma > 0$ and set

$$\chi(\zeta) = \frac{F}{\omega_0^2 - \zeta^2 - i\gamma\zeta}, \qquad \omega_0^2 - \omega^2 = \frac{a}{b}\gamma\omega, \qquad F = \gamma\omega\frac{a^2 + b^2}{b} > 0. \tag{46}$$

Then $\chi(\omega) = a + ib$. Independent oscillators on the real principal axes preserve locality and reciprocity. This compatibility observation is not part of the formal main theorem and gives no bandwidth claim.

The theorem supplies no practical finite-size optimizer, quantitative universal resource cost, common positive loss floor, or all-isotropic realization. Aperture and absorber thickness can grow, and the final isotropic loss can shrink. The fixed-resource and quartic small-object questions require different quantifiers and remain separate. No numerical simulation or experimental data are needed for the existence proof, and none is presented as a continuum accuracy certificate here.

10 Formal verification correspondence

The frozen Lean 4 repository is identified by commit

`686402a6c2d63820828729cc7ab02504632e3fe7`.

The principal declaration and its type are

```
OQ7Prescribed.theorem11_complete
:  PhysicalTheorem11Statement.
```

The source is `PrescribedExtinction/PhysicalTheorem11.lean`. The first conjunct preserves the existence statement `Theorem11Statement`. Further conjuncts prove both prescribed-extinction suprema, well-posedness almost everywhere for every admissible two-layer body, the full physical optical certificates, reciprocal extinction, universal bounds, and strict nonattainment.

The correspondence is literal at the level of the main class and powers. `TwoLayerBody` uses two positive finite thicknesses and a finite positive square side length. `MaterialAdmissible` imposes transpose symmetry and the same Hermitian quadratic-form bounds as (2). `ScatteringState` contains actual weak Maxwell fields, the uniform outgoing condition, and an L^2 compact polarization. Its test class is C_c^∞. The formal supremum sets range over every such admissible body and state pair at the prescribed extinction, rather than only the constructed family. Physical sphere flux is proved for the canonical radiating representative.

The machine proof of universal strictness also uses strictly positive local absorption at positive extinction, while Lemma 4.2 uses the far-field optical theorem. These are compatible proofs of the same conclusion within the stated class. Supplement S1: theorem correspondence distinguishes direct declarations, mathematical consequences, and exposition that is not separately formalized. The Weyl-fiber argument in Section 7 and the Lean localization proof, which uses physical-space and L^2 lemmas, have different organizations. Equality of theorem statements does not require identical proof organization.

The final AI-assisted independent audit of the frozen commit above reports PASS for Lean kernel validity, semantic fidelity to the original Theorem 1.1 and Lemma 1.2, Maxwell correctness within the formal model, both exact suprema and strict nonattainment, and closure of the preceding audit's F1–F4. These original statements correspond to Theorem 4.1 and Lemma 4.2 here; the version comparison is supplied with the release. The final theorem uses only `propext`, `Classical.choice`, and `Quot.sound`.

The auditor additionally proved `auditor_extinction_pos` and unconditional strictness for admissible bodies; those audit theorems are preserved separately, without changing the frozen main theorem. The final audit report, programs, and logs accompany this release.

The audit reports a fresh empty-environment replay of the `PhysicalTheorem11` import closure, using the same Lean kernel implementation. This is distinct from an independently implemented

checker, which was not used. Supplement S3: frozen verification manifest retains its historical export-time pending-review field; the completed final audit supersedes that status. Supplement S4: audit provenance records both the preceding and final reviews. The audit establishes the main theorem's statement correspondence, not verification of every intermediate statement or this manuscript's reorganized proof presentation.

Kernel checking verifies the encoded implication. The completed AI-assisted semantic review assesses its correspondence with the stated Maxwell model; it is distinct from human peer review, which is not established by these materials. Scientific priority, experimental realization, and finite-resource or bandwidth performance remain separate questions. Neither formal verification nor absence of a retrieved equivalent publication establishes novelty.

No experimental or simulation datasets are used. The proof code is distributed with this review manuscript as Supplement S2: frozen Lean source; its toolchain, dependencies, correspondence table, and checking instructions are listed in the supplement index. These are accompanying review materials; a public archival location remains unspecified.

11 Conclusion

The prescribed-extinction problem has a unit normalized supremum within finite compact two-layer reciprocal anisotropic dielectrics at one frequency and under arbitrary fixed positive upper material caps. The proof combines exact rotation factorization, uniform normalized aperture localization, and two finite-object continuity arguments. Each selected object retains the exact extinction target and strictly positive tensor loss, while strict optical-theorem nonattainment prevents equality for an individual body. This gives a precise resource-unrestricted answer to one formulation of OQ7; finite-resource optima and bandwidth remain separate questions.

A Entire outgoing weak vacuum uniqueness

This appendix supplies the converse-representation step without assuming classical traces of arbitrary weak representatives. Let $E, \mathbf{H} \in H_{\mathrm{loc}}(\mathrm{curl})$ solve the vacuum equations (3) on all of $\mathbb{R}^3$ and satisfy (4). Choose a smooth radial cutoff η_R equal to one on $r \le R$ and zero on $r \ge 2R$, with $w_R = -\eta_R' \ge 0$ and $\int_R^{2R} w_R \, \mathrm{d}r = 1$. Weighted weak Green identities follow by local mollification, compact integration by parts, and passage to the limit in products of local L^2 fields. The constant-coefficient curls commute with mollification; the products converge in L^1. In particular, the real Poynting identity gives

$$\int_{\mathbb{R}^3} w_R \Re\{E \cdot \overline{\mathbf{H} \times \mathbf{n}}\} \, \mathrm{d}\mathbf{x} = 0, \quad \mathbf{n} = \mathbf{x}/|\mathbf{x}|. \tag{47}$$

The pointwise norm-square identity and outgoing condition yield

$$\int w_R(\|E\|^2 + \|\mathbf{H} \times \mathbf{n}\|^2) = \int w_R \|\mathbf{H} \times \mathbf{n} - E\|^2 = o(1), \tag{48}$$

since $\int w_R/r^2 \, \mathrm{d}\mathbf{x} = 4\pi$.

For an arbitrary $q \in C_c^\infty(\mathbb{R}^3)^3$, let $(E_q, \mathbf{H}_q)$ be its outgoing volume fields, so $\nabla \times \mathbf{H}_q = -ik(E_q + q)$. These smooth fields satisfy $E_q = O(r^{-1})$ and $\mathbf{H}_q \times \mathbf{n} - E_q = O(r^{-2})$ outside the source support. The weak bilinear Lorentz identity, derived by the same compact product argument, is

$$ik \int \eta_R E \cdot q \, \mathrm{d}\mathbf{x} = \int w_R \{E \cdot (\mathbf{H}_q \times \mathbf{n}) - E_q \cdot (\mathbf{H} \times \mathbf{n})\} \, \mathrm{d}\mathbf{x}. \tag{49}$$

Rewrite the integrand as $E\cdot(\mathbf{H}_q\times\mathbf{n}-E_q)-E_q\cdot(\mathbf{H}\times\mathbf{n}-E)$. Cauchy–Schwarz and (48) bound the first term by $o(1)O(R^{-1})$, using $\int w_R/r^4\,\mathrm{d}\mathbf{x}\le 4\pi/R^2$. The second tends to zero by $E_q=O(r^{-1})$, uniform outgoing decay, and $\int w_R/r^2\,\mathrm{d}\mathbf{x}=4\pi$. For large R, $\eta_R=1$ on the fixed source support, so $\int E\cdot q=0$ for every smooth compact vector test. Distributional injectivity gives $E=0$ almost everywhere, and its curl equation gives $\mathbf{H}=0$ almost everywhere.

For a general weak scattering state, subtract the compact-source reconstruction with $p=\chi E$. The difference is an entire outgoing vacuum pair and hence vanishes. This proves the converse representation before applying constitutive volume-equation uniqueness. It also explains why equality of arbitrary raw weak representatives is stated almost everywhere, while the smooth exterior representative carries the pointwise radiation and flux limits.

B Nonsingular normalized planar amplitudes

The reference planes are $L:z=0$ and $R:z=L=h+d$. For b^+ and b^- the incident y amplitude is one at the left and right planes, respectively. The latter convention multiplies f_- by e^{ikL}, an overall unit-modulus phase that does not change powers. Outgoing amplitudes are read at these planes. Let $s=\sin\theta$, $c=\cos\theta$, $v_A=e^{ikh}$, $v_W=e^{ikd}$, $\Delta=t_W-v_W$, and $D=1-r_Ar_Wc^2$. With output order (Lx,Ly,Rx,Ry), subtract $v_Av_W\mathbf{e}_y$ from the transmitted amplitude and divide the scattered vector by s. Cascade matching gives

$$b^+=\frac{1}{D}\begin{pmatrix} r_Wv_At_Ac\\ r_Wv_A^2s\\ cv_A(\Delta+v_Wr_Ar_W)\\ sv_A\Delta\end{pmatrix}, \tag{50}$$

$$b^-=\begin{pmatrix} ct_A\Delta/D\\ sv_A\Delta/D\\ c\{r_A\Delta(v_W+\Delta c^2)/D+r_W\}\\ s\{r_A\Delta^2c^2/D+r_W\}\end{pmatrix}. \tag{51}$$

The scalar amplitudes are obtained by matching E,E' at the two boundaries of each layer; the common denominator sums the reflected x-channel feedback $r_Ar_Wc^2$. All expressions extend continuously at $s=0$ near $\delta=0$, where $D=1$. The normalized powers are $C_\infty^\pm=\|b^\pm\|^2$ and $B_\infty^\pm=Y_\infty-C_\infty^\pm$. Substituting $r_W=0,t_W=1,v_W=-1$ gives $C_\infty^+=4$ and $C_\infty^-=4(s^2+c^2(|r_A|^2+|t_A|^2))$, recovering Lemma 6.1 directly.

An operator argument independently justifies this regular limit. At $\delta=0$ the active scalar susceptibility stays nonzero. The projected transverse planar Green kernel on the finite interval is Hilbert–Schmidt, so $\alpha^{-1}-B_{\mathbf{0},\theta}$ is Fredholm of index zero. A nullvector reconstructs a source-free outgoing passive one-dimensional field. Its flux identity forces both outgoing amplitudes to vanish; exterior E,E' are therefore zero at an interface. Uniqueness of the finite-layer ordinary differential equation makes the field and current zero. The inverse exists, and operator-norm perturbation gives joint continuity in (δ,θ) near $(0,0)$ for the fixed auxiliary drives. No inverse estimate uniform as $\delta\downarrow 0$ is used in the aperture theorem.

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